Questions

1. A laboratory mixing vessel initially contains 4 litres of solution in which 4 mg of dye is dissolved.

A wash solution enters the vessel at a constant rate of 300 ml per minute.

At the same time, well-mixed liquid leaves the vessel at a constant rate of 200 ml per minute.

The concentration of dye in the liquid entering the vessel, at time t minutes after pumping begins, is modelled by $2e^{-0.1t}$ mg per litre.

Let m be the amount of dye, in milligrams, in the vessel at time t minutes.

(a) Show that the volume of liquid in the vessel is $(4 + 0.1t)$ litres, and hence show that m satisfies

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.6e^{-0.1t} - \frac{2m}{40 + t}$$ **[4]**

(b) Solve this differential equation, given that $m = 4$ when $t = 0$, to find an expression for m in terms of t **[8]**

(c) After 20 minutes, the concentration of dye in the vessel was measured as 0.82 mg per litre. Assess the model in light of this information, giving a reason for your answer **[2]**

Solution

(a) The liquid enters at 300 ml/min and leaves at 200 ml/min, so the net increase in volume is

$$300 - 200 = 100 \text{ ml/min} = 0.1 \text{ L/min}$$

Hence, starting from 4 L, the volume after t minutes is

$$V(t) = 4 + 0.1t$$

Now form a rate equation for the dye.
The dye enters with concentration $2e^{-0.1t}$ mg/L and flow rate 0.3 L/min, so the rate of dye entering is

$$0.3 \times 2e^{-0.1t} = 0.6e^{-0.1t} \text{ mg/min}$$

The concentration of dye in the vessel at time t is

$$\frac{m}{4 + 0.1t} \text{ mg/L}$$

Since liquid leaves at 0.2 L/min, the rate of dye leaving is

$$0.2\left(\frac{m}{4 + 0.1t}\right)$$

Simplifying,

$$0.2\left(\frac{m}{4 + 0.1t}\right) = \frac{0.2m}{4 + 0.1t} = \frac{2m}{40 + t}$$

Therefore,

$$\begin{aligned}\frac{\mathrm{d}m}{\mathrm{d}t} &= \text{rate in} - \text{rate out}\\ &= 0.6e^{-0.1t} - \frac{2m}{40 + t}\end{aligned}$$

So

$$V(t) = 4 + 0.1t \text{ litres}, \qquad \frac{\mathrm{d}m}{\mathrm{d}t} = 0.6e^{-0.1t} - \frac{2m}{40 + t}$$

(b) We solve

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{2m}{40+t} = 0.6e^{-0.1t}$$

This is a linear differential equation.

The integrating factor is

$$\begin{aligned}\mu(t) &= \exp\left(\int \frac{2}{40+t}\,\mathrm{d}t\right)\\ &= \exp\left(2\ln(40+t)\right)\\ &= (40+t)^2\end{aligned}$$

Multiplying the differential equation by $(40+t)^2$,

$$(40+t)^2\frac{\mathrm{d}m}{\mathrm{d}t} + (40+t)^2\frac{2m}{40+t} = 0.6(40+t)^2e^{-0.1t}$$

So

$$\frac{\mathrm{d}}{\mathrm{d}t}\left((40+t)^2m\right) = 0.6(40+t)^2e^{-0.1t}$$

Integrating both sides,

$$(40+t)^2m = \int 0.6(40+t)^2e^{-0.1t}\,\mathrm{d}t$$

Expand $(40+t)^2$:

$$(40+t)^2 = t^2 + 80t + 1600$$

So

$$(40+t)^2m = \int 0.6(t^2+80t+1600)e^{-0.1t}\,\mathrm{d}t$$

Let

$$I = \int 0.6(t^2+80t+1600)e^{-0.1t}\,\mathrm{d}t$$

and use integration by parts with

$$u = t^2+80t+1600, \qquad \mathrm{d}v = 0.6e^{-0.1t}\,\mathrm{d}t$$

Then

$$\mathrm{d}u = (2t+80)\,\mathrm{d}t, \qquad v = \int 0.6e^{-0.1t}\,\mathrm{d}t = -6e^{-0.1t}$$

So

$$\begin{aligned}I &= uv - \int v\,\mathrm{d}u\\ &= -6(t^2+80t+1600)e^{-0.1t} + 6\int(2t+80)e^{-0.1t}\,\mathrm{d}t\\ &= -6(t^2+80t+1600)e^{-0.1t} + 12\int(t+40)e^{-0.1t}\,\mathrm{d}t\end{aligned}$$

Now let

$$J = \int (t+40)e^{-0.1t}\,\mathrm{d}t$$

Again integrating by parts, take

$$u = t+40, \qquad \mathrm{d}v = e^{-0.1t}\,\mathrm{d}t$$

Then

$$\mathrm{d}u = \mathrm{d}t, \qquad v = \int e^{-0.1t}\,\mathrm{d}t = -10e^{-0.1t}$$

Hence

$$\begin{aligned} J &= uv - \int v\,\mathrm{d}u \\ &= -10(t+40)e^{-0.1t} + 10\int e^{-0.1t}\,\mathrm{d}t \\ &= -10(t+40)e^{-0.1t} - 100e^{-0.1t} + K \\ &= -10(t+50)e^{-0.1t} + K \end{aligned}$$

Substituting this into the expression for I,

$$\begin{aligned} I &= -6(t^2+80t+1600)e^{-0.1t} + 12\big(-10(t+50)e^{-0.1t}\big) + K \\ &= -6e^{-0.1t}\big(t^2+80t+1600+20t+1000\big) + K \\ &= -6e^{-0.1t}(t^2+100t+2600) + K \end{aligned}$$

Therefore,

$$\int 0.6(t^2+80t+1600)e^{-0.1t}\ \mathrm{d}t = -6e^{-0.1t}(t^2+100t+2600) + K$$

where K is a constant.

So

$$(40+t)^2 m = -6e^{-0.1t}(t^2+100t+2600) + K$$

Now use the initial condition $m = 4$ when $t = 0$:

$$\begin{aligned} (40)^2(4) &= -6e^0(0^2+100(0)+2600) + K \\ 6400 &= -6(2600) + K \\ 6400 &= -15600 + K \end{aligned}$$

Thus

$$K = 22000$$

Hence

$$(40+t)^2 m = 22000 - 6e^{-0.1t}(t^2+100t+2600)$$

and so

$$m = \frac{22000 - 6e^{-0.1t}(t^2+100t+2600)}{(40+t)^2}$$

Therefore, the amount of dye is

$$m(t) = \frac{22000 - 6e^{-0.1t}(t^2+100t+2600)}{(40+t)^2}\ \text{mg}$$

(c) After 20 minutes, the model gives volume

$$V(20) = 4 + 0.1(20) = 6\ \text{L}$$

Using the expression for m,

$$\begin{aligned} m(20) &= \frac{22000 - 6e^{-2}(20^2+100(20)+2600)}{(40+20)^2} \\ &= \frac{22000 - 6e^{-2}(400+2000+2600)}{60^2} \\ &= \frac{22000 - 30000e^{-2}}{3600} \\ &= \frac{55}{9} - \frac{25}{3}e^{-2} \end{aligned}$$

So the predicted concentration is

$$\begin{aligned}\frac{m(20)}{V(20)} &= \frac{1}{6}\left(\frac{55}{9} - \frac{25}{3}e^{-2}\right)\\ &= \frac{55}{54} - \frac{25}{18}e^{-2}\\ &\approx 0.831 \text{ mg/L}\end{aligned}$$

The measured concentration was 0.82 mg/L, which is very close to 0.831 mg/L.

So the model appears reasonable, since its prediction is consistent with the observed value.

2. A particle P, of mass $2\,\text{kg}$, moves in a straight line and has velocity $v\,\text{m}\,\text{s}^{-1}$.

At time t seconds, where $t \geq 0$, a variable force of $\dfrac{12t - 4v}{1+t}$ N acts on P.

There are no other forces acting on P. Initially, when $t = 0$, P has velocity $-5\,\text{m}\,\text{s}^{-1}$.

The motion of P can be modelled by the differential equation

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t.

(a) Find the functions $P(t)$ and $Q(t)$ **[2]**

(b) Using an integrating factor, determine the first time at which P is stationary according to the model **[8]**

Solution

(a) Since the mass of P is 2 kg and the only force acting is

$$\frac{12t - 4v}{1+t}$$

Newton's second law gives

$$2\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{12t - 4v}{1+t}$$

Rearranging into the form

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

gives

$$\begin{aligned} \frac{\mathrm{d}v}{\mathrm{d}t} &= \frac{12t - 4v}{2(1+t)} \\ &= \frac{6t - 2v}{1+t} \\ \frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2}{1+t}v &= \frac{6t}{1+t} \end{aligned}$$

Hence

$$P(t) = \frac{2}{1+t} \qquad \text{and} \qquad Q(t) = \frac{6t}{1+t}$$

(b) From part (a), the differential equation is

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2}{1+t}v = \frac{6t}{1+t}$$

The integrating factor is

$$\begin{aligned} \mu(t) &= e^{\int \frac{2}{1+t}\,\mathrm{d}t} \\ &= e^{2\ln(1+t)} \\ &= (1+t)^2 \end{aligned}$$

since $t \geq 0$, so $1 + t > 0$.
Multiplying the differential equation by $(1+t)^2$ gives

$$\begin{aligned} (1+t)^2\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{2(1+t)^2}{1+t}v &= \frac{6t(1+t)^2}{1+t} \\ (1+t)^2\frac{\mathrm{d}v}{\mathrm{d}t} + 2(1+t)v &= 6t(1+t) \end{aligned}$$

The left-hand side is now the derivative of $(1+t)^2 v$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big((1+t)^2 v\big) = 6t(1+t)$$

Integrating with respect to t,

$$\begin{aligned}(1+t)^2 v &= \int 6t(1+t)\,\mathrm{d}t \\ &= \int (6t+6t^2)\,\mathrm{d}t \\ &= 3t^2 + 2t^3 + C\end{aligned}$$

Use the initial condition $v = -5$ when $t = 0$:

$$\begin{aligned}(1+0)^2(-5) &= 3(0)^2 + 2(0)^3 + C \\ -5 &= C\end{aligned}$$

So

$$(1+t)^2 v = 3t^2 + 2t^3 - 5$$

and therefore

$$v = \frac{2t^3 + 3t^2 - 5}{(1+t)^2}$$

Factorising the numerator,

$$2t^3 + 3t^2 - 5 = (t-1)(2t^2 + 5t + 5)$$

so

$$v = \frac{(t-1)(2t^2+5t+5)}{(1+t)^2}$$

The particle is stationary when $v = 0$, so we solve

$$\frac{(t-1)(2t^2+5t+5)}{(1+t)^2} = 0$$

For $t \geq 0$, we have $(1+t)^2 > 0$, and also

$$2t^2 + 5t + 5 > 0$$

so the only way $v = 0$ is when

$$t - 1 = 0$$

Hence

$$t = 1$$

Therefore, the first time at which P is stationary is 1 s.

3. A brief flushing cycle in an industrial pipe is to be modelled. The flow rate of liquid, R litres per second, at time t seconds is measured from the instant the valve is opened. The cycle starts when $t = 0$ and there is initially no flow through the pipe. When $t = 1$, the flow rate is measured as $\pi e^{-1}\sin(1)$ litres per second.

While the cycle is occurring, R is modelled by the differential equation

$$\frac{\mathrm{d}R}{\mathrm{d}t} + (1 - \cot t)R = \frac{e^{-t}\sin t}{1+t^2}$$

(a) By using an integrating factor show that, according to the model, while the cycle is occurring,

$$R = e^{-t}\sin t\left(\tan^{-1} t + \frac{3\pi}{4}\right)$$ **[6]**

The cycle ends when there is again no flow through the pipe.

(b) Determine the length of time that the cycle lasts according to the model **[2]**

(c) Determine, according to the model, the rate of increase of the flow rate at the start of the cycle. Give your answer in exact form **[3]**

Solution

(a) For the cycle we consider $0 < t < \pi$, so $\sin t > 0$. The differential equation is

$$\frac{\mathrm{d}R}{\mathrm{d}t} + (1 - \cot t)R = \frac{e^{-t}\sin t}{1+t^2}$$

An integrating factor is

$$\begin{aligned}\mu(t) &= e^{\int(1-\cot t)\ \mathrm{d}t}\\ &= e^{\,t-\int \cot t\ \mathrm{d}t}\\ &= e^{\,t-\ln(\sin t)}\\ &= \frac{e^t}{\sin t}\end{aligned}$$

Multiplying the differential equation by $\dfrac{e^t}{\sin t}$ gives

$$\begin{aligned}\frac{e^t}{\sin t}\frac{\mathrm{d}R}{\mathrm{d}t} + \frac{e^t}{\sin t}(1-\cot t)R &= \frac{e^t}{\sin t}\cdot\frac{e^{-t}\sin t}{1+t^2}\\ &= \frac{1}{1+t^2}\end{aligned}$$

Now

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{e^t}{\sin t}\right) &= \frac{e^t}{\sin t} - \frac{e^t\cos t}{\sin^2 t}\\ &= \frac{e^t}{\sin t}\left(1 - \frac{\cos t}{\sin t}\right)\\ &= \frac{e^t}{\sin t}(1-\cot t)\end{aligned}$$

so the left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{e^t}{\sin t}R\right)$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{e^t}{\sin t}R\right) = \frac{1}{1+t^2}$$

Integrating:

$$\begin{aligned}\frac{e^t}{\sin t}R &= \int \frac{1}{1+t^2}\,\mathrm{d}t\\ &= \tan^{-1} t + C\end{aligned}$$

Therefore

$$R = e^{-t}\sin t\left(\tan^{-1} t + C\right)$$

Use the condition $R(1) = \pi e^{-1}\sin 1$:

$$\begin{aligned}\pi e^{-1}\sin 1 &= e^{-1}\sin 1\left(\tan^{-1} 1 + C\right)\\ &= e^{-1}\sin 1\left(\frac{\pi}{4} + C\right)\end{aligned}$$

Since $e^{-1}\sin 1 \neq 0$,

$$\begin{aligned}\pi &= \frac{\pi}{4} + C\\ C &= \frac{3\pi}{4}\end{aligned}$$

So

$$R = e^{-t}\sin t\left(\tan^{-1} t + \frac{3\pi}{4}\right)$$

Thus, according to the model,

$$R = e^{-t}\sin t\left(\tan^{-1} t + \frac{3\pi}{4}\right)$$

(b) The cycle ends when there is again no flow, so we set $R = 0$:

$$e^{-t}\sin t\left(\tan^{-1} t + \frac{3\pi}{4}\right) = 0$$

For $t \geq 0$, we have $e^{-t} > 0$ and

$$\tan^{-1} t + \frac{3\pi}{4} > 0$$

so the only way $R = 0$ is if

$$\sin t = 0$$

Hence

$$t = n\pi \qquad (n \in \mathbb{Z})$$

and the first time after $t = 0$ is

$$t = \pi$$

Therefore the cycle lasts π seconds.

(c) The rate of increase of the flow rate at the start is $R'(0)$.
From part (a),

$$R(t) = e^{-t}\sin t\left(\tan^{-1} t + \frac{3\pi}{4}\right).$$

Differentiate:

$$R'(t) = e^{-t}\left[\cos t\left(\tan^{-1} t + \frac{3\pi}{4}\right) + \frac{\sin t}{1+t^2} - \sin t\left(\tan^{-1} t + \frac{3\pi}{4}\right)\right].$$

Hence

$$R'(0) = 1\left[1 \cdot \frac{3\pi}{4} + 0 - 0\right] = \frac{3\pi}{4}.$$

So the rate of increase of the flow rate at the start is

$$\frac{3\pi}{4}\ \mathrm{L\,s^{-2}}.$$

4. A mixing tank initially contains 600 litres of pure water.

Brine flows into the tank at a constant rate of 20 litres per hour. Each litre of brine contains 3 grams of dissolved salt.

The well-stirred mixture is pumped out of the tank at a constant rate of 30 litres per hour.

It is assumed that the salt is instantly and uniformly mixed throughout the liquid.

Given that there are x grams of salt in the tank after t hours,

(a) show that, while the tank still contains liquid, the situation can be modelled by

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 60 - \frac{3x}{60 - t}$$ **[4]**

(b) Hence find the number of grams of salt in the tank after 30 hours **[5]**

(c) State one limitation of this model and explain briefly how the model could be refined **[1]**

Solution

(a) Let the volume of liquid in the tank after t hours be V litres.

Since liquid flows in at 20 litres/h and out at 30 litres/h, the net change in volume is

$$V = 600 + 20t - 30t = 600 - 10t = 10(60 - t)$$

This is valid while the tank still contains liquid, so $0 \leq t < 60$.

The rate at which salt enters is

$$20 \times 3 = 60 \text{ g/h}$$

The concentration of salt in the tank at time t is

$$\frac{x}{V} = \frac{x}{600 - 10t} \text{ g/litre}$$

Since the mixture leaves at 30 litres/h, the rate at which salt leaves is

$$30\left(\frac{x}{600 - 10t}\right) = \frac{30x}{600 - 10t} = \frac{3x}{60 - t} \text{ g/h}$$

Hence,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \text{rate in} - \text{rate out}$$

so

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 60 - \frac{3x}{60 - t}$$

Therefore, while the tank contains liquid,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 60 - \frac{3x}{60 - t}$$

(b) We solve

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{60 - t} = 60$$

with the initial condition $x(0) = 0$, since the tank initially contains pure water.

The integrating factor is

$$\begin{aligned}\mu(t) &= \exp\left(\int \frac{3}{60-t}\ \mathrm{d}t\right)\\ &= \exp\left(-3\ln(60-t)\right)\\ &= (60-t)^{-3}\end{aligned}$$

Multiplying the differential equation by $(60-t)^{-3}$ gives

$$(60-t)^{-3}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{60-t}(60-t)^{-3} = 60(60-t)^{-3}$$

The left-hand side is now the derivative of $x(60-t)^{-3}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[x(60-t)^{-3}\right] = 60(60-t)^{-3}$$

Integrating both sides with respect to t,

$$x(60-t)^{-3} = \int 60(60-t)^{-3}\ \mathrm{d}t + C$$

To evaluate the integral, let $u = 60-t$, so $\mathrm{d}u = -\mathrm{d}t$:

$$\begin{aligned}\int 60(60-t)^{-3}\ \mathrm{d}t &= -60\int u^{-3}\ \mathrm{d}u\\ &= -60\left(\frac{u^{-2}}{-2}\right)\\ &= 30u^{-2}\\ &= 30(60-t)^{-2}\end{aligned}$$

Therefore,

$$x(60-t)^{-3} = 30(60-t)^{-2} + C$$

Multiplying through by $(60-t)^3$,

$$x = 30(60-t) + C(60-t)^3$$

Use $x(0) = 0$:

$$\begin{aligned}0 &= 30(60) + C(60)^3\\ 0 &= 1800 + 216000C\\ C &= -\frac{1800}{216000} = -\frac{1}{120}\end{aligned}$$

So

$$x = 30(60-t) - \frac{(60-t)^3}{120}$$

At $t = 30$,

$$\begin{aligned}x(30) &= 30(60-30) - \frac{(60-30)^3}{120}\\ &= 30\cdot 30 - \frac{30^3}{120}\\ &= 900 - \frac{27000}{120}\\ &= 900 - 225\\ &= 675\end{aligned}$$

Hence the number of grams of salt in the tank after 30 hours is

675 g

(c) One limitation is that the model assumes the salt is mixed instantly and uniformly throughout the tank, which is not completely realistic.

A refinement would be to use a model with finite mixing time, for example a compartment model or a model where concentration varies in different parts of the tank.

5. The concentration C, in suitable units, of a pollutant in a reservoir at time t hours after a filtration system is switched on is to be modelled by a differential equation. Initially, the concentration is A, and its long-term value is 0.

(a) One simple model is to assume that the rate of decrease of concentration is directly proportional to C.

(i) Formulate a differential equation for this model **[1]**

(ii) Verify that $C = Ae^{-kt}$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition **[3]**

An alternative model uses the differential equation

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = R(t)$$

where $R(t)$ is a function of t.

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$1 + t^2$$ **[3]**

(c) Suppose that $R(t) = 0$.

(i) Show that

$$C = \frac{A}{1+t^2}$$ **[4]**

(ii) Find the time predicted by this model for the concentration to fall to half its initial value. Give your answer correct to the nearest minute **[2]**

(d) Now suppose that

$$R(t) = \frac{te^{-t}}{1+t^2}$$

Show that

$$C = \frac{A + 1 - (1+t)e^{-t}}{1+t^2}$$ **[6]**

It is found that the initial concentration is 5, and the concentration falls to half this value after 63 minutes.

(e) Determine which of the models proposed in parts (c) and (d) is more consistent with these data **[2]**

Solution

(a) (i) If the rate of decrease of concentration is directly proportional to C, then

$$-\frac{\mathrm{d}C}{\mathrm{d}t} \propto C$$

so, for some constant $k > 0$,

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC$$

Hence the differential equation is $\frac{\mathrm{d}C}{\mathrm{d}t} = -kC$.

(ii) Take

$$C = Ae^{-kt}$$

Then

$$\frac{\mathrm{d}C}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(Ae^{-kt}) = -kAe^{-kt}$$

Since $Ae^{-kt} = C$, this becomes

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC$$

so the differential equation is satisfied.
For the initial condition,

$$C(0) = Ae^{-k\cdot 0} = A$$

so the initial concentration is correct.
For the long-term condition, since $k > 0$,

$$\lim_{t\to\infty} C = \lim_{t\to\infty} Ae^{-kt} = A \lim_{t\to\infty} e^{-kt} = 0$$

Hence $C = Ae^{-kt}$ satisfies the differential equation, the initial condition and the long-term condition.

(b) For

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = R(t)$$

the integrating factor is

$$\mu(t) = e^{\int \frac{2t}{1+t^2}\,\mathrm{d}t}$$

To evaluate the integral, let

$$u = 1 + t^2 \qquad \Rightarrow \qquad \mathrm{d}u = 2t\,\mathrm{d}t$$

Then

$$\begin{aligned}\int \frac{2t}{1+t^2}\,\mathrm{d}t &= \int \frac{1}{u}\,\mathrm{d}u \\ &= \ln u \\ &= \ln(1+t^2)\end{aligned}$$

since $1 + t^2 > 0$.
Therefore

$$\mu(t) = e^{\ln(1+t^2)} = 1 + t^2$$

Hence an integrating factor is $1 + t^2$.

(c) (i) If $R(t) = 0$, the differential equation becomes

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = 0$$

Multiply through by the integrating factor $1 + t^2$:

$$(1+t^2)\frac{\mathrm{d}C}{\mathrm{d}t} + 2tC = 0$$

The left-hand side is the derivative of $(1+t^2)C$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\big((1+t^2)C\big) = 0$$

Integrating gives

$$(1+t^2)C = K$$

where K is a constant.
Use the initial condition $C(0) = A$:

$$(1+0^2)A = K$$

so $K = A$. Therefore

$$C = \frac{A}{1+t^2}$$

Hence

$$C = \frac{A}{1+t^2}$$

(ii) Half the initial concentration is $\frac{A}{2}$, so set

$$\frac{A}{1+t^2} = \frac{A}{2}$$

Cancelling A,

$$\frac{1}{1+t^2} = \frac{1}{2}$$

so

$$1+t^2 = 2$$
$$t^2 = 1$$

Since $t \geq 0$,

$$t = 1 \text{ hour}$$

Therefore the time is 60 minutes.

(d) If

$$R(t) = \frac{te^{-t}}{1+t^2}$$

then the differential equation is

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = \frac{te^{-t}}{1+t^2}$$

Multiply through by the integrating factor $1+t^2$:

$$(1+t^2)\frac{\mathrm{d}C}{\mathrm{d}t} + 2tC = te^{-t}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\big((1+t^2)C\big) = te^{-t}$$

Integrating,

$$(1+t^2)C = \int te^{-t}\,\mathrm{d}t$$

Evaluate the integral by parts. Let

$$u = t, \qquad \mathrm{d}v = e^{-t}\,\mathrm{d}t$$

Then

$$\mathrm{d}u = \mathrm{d}t, \qquad v = -e^{-t}$$

So

$$\begin{aligned}\int te^{-t}\,\mathrm{d}t &= uv - \int v\,\mathrm{d}u \\ &= t(-e^{-t}) - \int (-e^{-t})\,\mathrm{d}t \\ &= -te^{-t} + \int e^{-t}\,\mathrm{d}t \\ &= -te^{-t} - e^{-t} + K \\ &= -(1+t)e^{-t} + K\end{aligned}$$

Therefore

$$(1+t^2)C = -(1+t)e^{-t} + K$$

Use $C(0) = A$:

$$\begin{aligned}(1+0^2)A &= -(1+0)e^0 + K \\ A &= -1 + K\end{aligned}$$

so

$$K = A + 1$$

Hence

$$C = \frac{A + 1 - (1+t)e^{-t}}{1+t^2}$$

Also, using $\lim_{t\to\infty}(1+t)e^{-t} = 0$,

$$\lim_{t\to\infty} C = \lim_{t\to\infty} \frac{A + 1 - (1+t)e^{-t}}{1+t^2} = 0$$

so this model also has the correct long-term value.

(e) From part (c), model (c) predicts the concentration falls to half its initial value after 60 minutes. For model (d), with $A = 5$,

$$C = \frac{6 - (1+t)e^{-t}}{1+t^2}$$

Half the initial concentration is 2.5, so solve

$$\frac{6 - (1+t)e^{-t}}{1+t^2} = 2.5$$

Using a calculator gives

$$t \approx 1.055 \text{ hours}$$

which is

$$1.055 \times 60 \approx 63.3 \text{ minutes}$$

This is much closer to the observed 63 minutes than the 60 minutes from model (c). Therefore model (d) is more consistent with the data.

6. A processing tank initially contains 15 litres of water in which 6 mg of dye is dissolved.

A dye solution enters the tank at a constant rate of 200 ml per minute. The concentration of dye in this incoming solution, at time t minutes after the flow begins, is modelled by $(4 + e^{-0.1t})$ mg per litre.

At the same time, pure water enters through a second pipe at a constant rate of 100 ml per minute. The liquid in the tank is continually mixed and leaves the tank at a constant rate of 300 ml per minute.

Let m be the amount of dye, in milligrams, in the tank at time t minutes after the flow begins.

(a) (i) Write down the volume of liquid in the tank at time t

(ii) Hence show that m satisfies the differential equation

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50}$$ **[4]**

(b) By solving the differential equation, find an expression for m in terms of t **[8]**

After 30 minutes, the concentration of dye in the tank was measured as 1.52 mg per litre.

(c) Assess the model in light of this information, giving a reason for your answer **[2]**

Solution

(a) (i) Convert the flow rates to litres per minute:

$$200 \text{ ml min}^{-1} = 0.2 \text{ L min}^{-1}, \qquad 100 \text{ ml min}^{-1} = 0.1 \text{ L min}^{-1}, \qquad 300 \text{ ml min}^{-1} = 0.3 \text{ L min}^{-1}$$

So the net rate of change of volume is

$$0.2 + 0.1 - 0.3 = 0 \text{ L min}^{-1}$$

Hence the volume stays constant at its initial value of 15 litres:

$$V(t) = 15$$

So the volume of liquid in the tank at time t is 15 litres.

(ii) The rate at which dye enters through the first pipe is

$$0.2\left(4 + e^{-0.1t}\right) \text{ mg min}^{-1}$$

so

$$\text{rate in} = 0.8 + 0.2e^{-0.1t}$$

The second pipe contains pure water, so it contributes no dye.
Since the tank is well mixed and has volume 15 L, the concentration of dye in the tank is

$$\frac{m}{15} \text{ mg L}^{-1}$$

Therefore the rate at which dye leaves the tank is

$$0.3 \times \frac{m}{15} = \frac{0.3m}{15} = \frac{m}{50} \text{ mg min}^{-1}$$

Hence

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \text{rate in} - \text{rate out}$$

so

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50}$$

as required.

(b) Rearranging the differential equation into linear form gives

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{1}{50}m = 0.8 + 0.2e^{-0.1t}$$

The integrating factor is

$$\exp\left(\int \frac{1}{50}\,\mathrm{d}t\right) = e^{t/50}$$

Multiply the differential equation by $e^{t/50}$:

$$\begin{aligned} e^{t/50}\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{1}{50}e^{t/50}m &= 0.8e^{t/50} + 0.2e^{-0.1t}e^{t/50} \\ &= 0.8e^{t/50} + 0.2e^{-0.08t} \end{aligned}$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(me^{t/50}\right)$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(me^{t/50}\right) = 0.8e^{t/50} + 0.2e^{-0.08t}$$

Integrating both sides with respect to t:

$$\begin{aligned} me^{t/50} &= \int 0.8e^{t/50}\,\mathrm{d}t + \int 0.2e^{-0.08t}\,\mathrm{d}t + C \\ &= 0.8(50)e^{t/50} + 0.2\left(\frac{1}{-0.08}\right)e^{-0.08t} + C \\ &= 40e^{t/50} - 2.5e^{-0.08t} + C \end{aligned}$$

Divide through by $e^{t/50}$:

$$\begin{aligned} m &= 40 - 2.5e^{-0.08t}e^{-t/50} + Ce^{-t/50} \\ &= 40 - 2.5e^{-0.1t} + Ce^{-0.02t} \end{aligned}$$

Initially there are 6 mg of dye in the tank, so $m(0) = 6$. Substituting $t = 0$ gives

$$\begin{aligned} 6 &= 40 - 2.5 + C \\ 6 &= 37.5 + C \\ C &= -31.5 \end{aligned}$$

Therefore

$$m = 40 - 2.5e^{-0.1t} - 31.5e^{-0.02t}$$

So the required expression is $m = 40 - 2.5e^{-0.1t} - 31.5e^{-0.02t}$.

(c) The model predicts that at $t = 30$ minutes,

$$m(30) = 40 - 2.5e^{-3} - 31.5e^{-0.6}$$

so the concentration is

$$\frac{m(30)}{15} = \frac{40 - 2.5e^{-3} - 31.5e^{-0.6}}{15} \approx 1.51 \text{ mg L}^{-1}$$

The measured concentration was 1.52 mg L^{-1}, which is very close to the model prediction. Therefore the model appears to be reasonable.

7. A particle P, of mass $2\,\mathrm{kg}$, moves in a straight line and has velocity $v\,\mathrm{m\,s^{-1}}$.

At time t seconds, where $t \geq 0$, a variable force of $-2t(v+3)\,\mathrm{N}$ acts on P. There are no other forces acting on P. Initially, when $t = 0$, P has velocity $5\,\mathrm{m\,s^{-1}}$.

The motion of P can be modelled by the differential equation

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t.

(a) Find the functions $P(t)$ and $Q(t)$ **[2]**

(b) Using an integrating factor, determine the first time at which P is stationary according to the model **[8]**

Solution

(a) By Newton's second law, resultant force = mass × acceleration, so

$$2\frac{\mathrm{d}v}{\mathrm{d}t} = -2t(v+3)$$

Expanding and dividing by 2 gives

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -tv - 3t$$

Hence

$$\frac{\mathrm{d}v}{\mathrm{d}t} + tv = -3t$$

Therefore, $P(t) = t$ and $Q(t) = -3t$.

(b) From part (a), the differential equation is

$$\frac{\mathrm{d}v}{\mathrm{d}t} + tv = -3t$$

The integrating factor is

$$e^{\int t\,\mathrm{d}t} = e^{t^2/2}$$

Multiplying the differential equation by $e^{t^2/2}$ gives

$$e^{t^2/2}\frac{\mathrm{d}v}{\mathrm{d}t} + tve^{t^2/2} = -3te^{t^2/2}$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t^2/2}\right)$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t^2/2}\right) = -3te^{t^2/2}$$

Integrating with respect to t,

$$\begin{aligned} ve^{t^2/2} &= \int -3te^{t^2/2}\,\mathrm{d}t \\ &= -3e^{t^2/2} + C \end{aligned}$$

Therefore

$$v = -3 + Ce^{-t^2/2}$$

Using the initial condition $v = 5$ when $t = 0$,

$$\begin{aligned} 5 &= -3 + Ce^0 \\ 5 &= -3 + C \\ C &= 8 \end{aligned}$$

Hence
$$v = -3 + 8e^{-t^2/2}$$
For P to be stationary, $v = 0$. So
$$\begin{aligned} 0 &= -3 + 8e^{-t^2/2} \\ 8e^{-t^2/2} &= 3 \\ e^{-t^2/2} &= \frac{3}{8} \end{aligned}$$
Taking natural logarithms,
$$\begin{aligned} -\frac{t^2}{2} &= \ln\left(\frac{3}{8}\right) \\ \frac{t^2}{2} &= \ln\left(\frac{8}{3}\right) \\ t^2 &= 2\ln\left(\frac{8}{3}\right) \end{aligned}$$
Since $t \geq 0$, the first time is
$$t = \sqrt{2\ln\left(\frac{8}{3}\right)}$$
Numerically,
$$t \approx 1.40 \text{ s}$$
Therefore, the first time at which P is stationary is $\sqrt{2\ln\left(\frac{8}{3}\right)}$ s, approximately 1.40 s.

8. A small rescue boat moves in a straight line on calm water. It starts from rest, is driven by its engine for 5 seconds and then, for the next 5 seconds, the engine is cut and a drogue is deployed. The boat comes to rest at the end of the 10 seconds. The total mass of the boat and crew is m kg, and at time t seconds, for $0 \leq t \leq 10$, the boat's velocity is $v \, \mathrm{m\,s^{-1}}$.
A resistance to motion, modelled by a force of magnitude $0.2mv$ N, acts on the boat throughout the 10 seconds.

(a) Explain why modelling the resistance to motion in this way is likely to be more realistic than assuming this force is constant **[1]**

(b) During the drogue phase of the motion, for $5 \leq t \leq 10$, the drogue exerts an additional constant resistance force of magnitude m N and the engine provides no driving force. Show that, for $5 \leq t \leq 10$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = -1$$ **[1]**

(c) (i) Solve the differential equation in part (b) **[4]**

(ii) Hence find the velocity of the boat when $t = 5$ **[1]**

(d) During the powered phase ($0 \leq t \leq 5$), the propeller provides a driving force of magnitude directly proportional to t^2. Show that, for $0 \leq t \leq 5$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = \lambda t^2$$ **[1]**

where λ is a positive constant

(e) (i) Show by integration that, for $0 \leq t \leq 5$,

$$v = 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right)$$ **[6]**

(ii) Hence find λ **[2]**

(f) Find the total distance, to the nearest metre, travelled by the boat during the motion **[6]**

Solution

(a) A resistive force depending on v is more realistic because water resistance increases as speed increases, and when $v = 0$ this model gives zero resistance. A constant resistive force would not reflect either of these features.

(b) During $5 \leq t \leq 10$, the only horizontal forces are the two resistances:

$$0.2mv \quad \text{and} \quad m$$

both acting opposite to the motion. Taking the direction of motion as positive and using Newton's second law,

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -0.2mv - m$$

Dividing by m gives

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = -1$$

as required.

(c) (i) For $5 \leq t \leq 10$, we solve

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}v = -1$$

The integrating factor is

$$e^{\int \frac{1}{5}\,\mathrm{d}t} = e^{t/5}$$

Multiplying the differential equation by $e^{t/5}$,

$$\begin{aligned} e^{t/5}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}e^{t/5}v &= -e^{t/5} \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/5}\right) &= -e^{t/5} \end{aligned}$$

Integrating,

$$\begin{aligned} ve^{t/5} &= \int -e^{t/5}\,\mathrm{d}t \\ &= -5e^{t/5} + C \end{aligned}$$

Hence

$$v = -5 + Ce^{-t/5}$$

The boat comes to rest at $t = 10$, so $v = 0$ when $t = 10$:

$$\begin{aligned} 0 &= -5 + Ce^{-2} \\ Ce^{-2} &= 5 \\ C &= 5e^2 \end{aligned}$$

Therefore

$$v = 5e^2e^{-t/5} - 5 = 5\left(e^{2-t/5} - 1\right)$$

So the solution is $v = 5\left(e^{2-t/5} - 1\right)$ for $5 \leq t \leq 10$.

(ii) Substituting $t = 5$ into the result from part (i),

$$\begin{aligned} v(5) &= 5\left(e^{2-1} - 1\right) \\ &= 5(e - 1) \end{aligned}$$

Hence the velocity when $t = 5$ is $5(e - 1)\,\mathrm{m\,s^{-1}}$.

(d) Let the driving force during $0 \leq t \leq 5$ be λmt^2, where $\lambda > 0$ is a constant. Then Newton's second law gives

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = \lambda mt^2 - 0.2mv$$

Dividing by m,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = \lambda t^2$$

as required.

(e) (i) We solve

$$\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}v = \lambda t^2$$

The integrating factor is again

$$e^{t/5}$$

Multiplying through by $e^{t/5}$,

$$\begin{aligned} e^{t/5}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{5}e^{t/5}v &= \lambda t^2e^{t/5} \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/5}\right) &= \lambda t^2e^{t/5} \end{aligned}$$

So

$$ve^{t/5} = \lambda\int t^2e^{t/5}\,\mathrm{d}t + C$$

Let

$$I = \int t^2e^{t/5}\,\mathrm{d}t$$

Integrating by parts with $u = t^2$ and $\mathrm{d}w = e^{t/5}\,\mathrm{d}t$,

$$\begin{aligned} I &= 5t^2e^{t/5} - \int 10te^{t/5}\,\mathrm{d}t \\ &= 5t^2e^{t/5} - 10\int te^{t/5}\,\mathrm{d}t \end{aligned}$$

Now let

$$J = \int te^{t/5}\,\mathrm{d}t$$

Again integrating by parts,

$$\begin{aligned} J &= 5te^{t/5} - \int 5e^{t/5}\,\mathrm{d}t \\ &= 5te^{t/5} - 25e^{t/5} \end{aligned}$$

Hence

$$\begin{aligned} I &= 5t^2e^{t/5} - 10\left(5te^{t/5} - 25e^{t/5}\right) \\ &= 5t^2e^{t/5} - 50te^{t/5} + 250e^{t/5} \\ &= 5e^{t/5}\left(t^2 - 10t + 50\right) \end{aligned}$$

Therefore

$$\begin{aligned} ve^{t/5} &= \lambda\left[5e^{t/5}\left(t^2 - 10t + 50\right)\right] + C \\ v &= 5\lambda\left(t^2 - 10t + 50\right) + Ce^{-t/5} \end{aligned}$$

The boat starts from rest, so $v = 0$ when $t = 0$:

$$\begin{aligned} 0 &= 5\lambda(50) + C \\ C &= -250\lambda \end{aligned}$$

Hence

$$\begin{aligned} v &= 5\lambda\left(t^2 - 10t + 50\right) - 250\lambda e^{-t/5} \\ &= 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right) \end{aligned}$$

Thus, for $0 \le t \le 5$,

$$v = 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right)$$

(ii) At $t = 5$, the expressions for v from parts (c) and (e)(i) must be equal. From part (e)(i),

$$\begin{aligned} v(5) &= 5\lambda\left(25 - 50 + 50 - 50e^{-1}\right) \\ &= 5\lambda\left(25 - \frac{50}{e}\right) \end{aligned}$$

From part (c)(ii),

$$v(5) = 5(e - 1)$$

Equating these,

$$\begin{aligned} 5\lambda\left(25 - \frac{50}{e}\right) &= 5(e - 1) \\ \lambda\left(25 - \frac{50}{e}\right) &= e - 1 \\ \lambda\left(\frac{25e - 50}{e}\right) &= e - 1 \\ \lambda\left(\frac{25(e - 2)}{e}\right) &= e - 1 \end{aligned}$$

So

$$\lambda = \frac{e(e-1)}{25(e-2)}$$

Therefore

$$\lambda = \frac{e(e-1)}{25(e-2)} \approx 0.260$$

(f) Since the boat moves forwards throughout the motion, the total distance is

$$s = \int_0^{10} v\,\mathrm{d}t = \int_0^5 v\,\mathrm{d}t + \int_5^{10} v\,\mathrm{d}t$$

For the powered phase,

$$\begin{aligned} s_1 &= \int_0^5 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right)\mathrm{d}t \\ &= 5\lambda\left[\frac{t^3}{3} - 5t^2 + 50t + 250e^{-t/5}\right]_0^5 \\ &= 5\lambda\left(\frac{125}{3} - 125 + 250 + \frac{250}{e} - 250\right) \\ &= 5\lambda\left(\frac{125}{3} - 125 + \frac{250}{e}\right) \\ &= 5\lambda\left(-\frac{250}{3} + \frac{250}{e}\right) \\ &= 1250\lambda\left(\frac{3-e}{3e}\right) \end{aligned}$$

Substituting $\lambda = \dfrac{e(e-1)}{25(e-2)}$,

$$\begin{aligned} s_1 &= 1250\left(\frac{e(e-1)}{25(e-2)}\right)\left(\frac{3-e}{3e}\right) \\ &= \frac{50(e-1)(3-e)}{3(e-2)} \end{aligned}$$

So

$$s_1 \approx 11.2 \text{ m}$$

For the drogue phase,

$$\begin{aligned} s_2 &= \int_5^{10} 5\left(e^{2-t/5} - 1\right)\mathrm{d}t \\ &= 5\left[\int_5^{10} e^{2-t/5}\,\mathrm{d}t - \int_5^{10} 1\,\mathrm{d}t\right] \\ &= 5\left[-5e^{2-t/5} - t\right]_5^{10} \\ &= 5\left((-5-10) - (-5e-5)\right) \\ &= 25(e-2) \end{aligned}$$

So

$$s_2 \approx 17.9 \text{ m}$$

Therefore the total distance is

$$\begin{aligned} s &= s_1 + s_2 \\ &= \frac{50(e-1)(3-e)}{3(e-2)} + 25(e-2) \\ &\approx 11.2 + 17.9 \\ &\approx 29.2 \end{aligned}$$

Hence the total distance travelled, to the nearest metre, is 29 m.

9. A winch raises a lift cage vertically. The cage starts from rest and comes to rest again 8 seconds later. The combined mass of the cage and its load is m kg, and at time t seconds, for $0 \leq t \leq 8$, the cage has upward velocity $v\,\mathrm{m\,s^{-1}}$.
A resistance to the motion, modelled by a force of magnitude $0.25mv$ N, acts on the cage throughout the 8 seconds.

(a) Explain why modelling the resistance to the motion in this way is likely to be more realistic than assuming this force is constant **[1]**

(b) During the final stage of the motion, for $4 \leq t \leq 8$, the winch provides a constant upward tension of magnitude $0.75mg$ N. Show that, for $4 \leq t \leq 8$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = -\frac{1}{4}g$$

[1]

(c) (i) Solve the differential equation in part (b) **[5]**

(ii) Hence find the velocity of the cage when $t = 4$ **[1]**

(d) During the first stage of the motion, for $0 \leq t \leq 4$, the winch provides an upward tension of magnitude $m(g + \lambda t)$ N, where λ is a positive constant. Show that, for $0 \leq t \leq 4$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = \lambda t$$

[1]

(e) (i) Show by integration that, for $0 \leq t \leq 4$,

$$v = 4\lambda\left(t - 4 + 4e^{-t/4}\right)$$

[5]

(ii) Hence find λ **[2]**

(f) Find the total distance, to the nearest metre, through which the cage rises during the motion **[6]**

Solution

(a) A resistive force usually depends on the speed of the moving object. When the cage moves faster, the resistance is greater, so modelling it as proportional to v is more realistic than taking it to be constant.

(b) For $4 \leq t \leq 8$, taking upwards as positive:

$$\text{tension} = 0.75mg, \qquad \text{weight} = mg, \qquad \text{resistance} = 0.25mv$$

Using Newton's second law,

$$\begin{aligned} m\frac{\mathrm{d}v}{\mathrm{d}t} &= 0.75mg - mg - 0.25mv \\ &= -0.25mg - 0.25mv \end{aligned}$$

Dividing by m,

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -0.25g - 0.25v$$

Hence

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = -\frac{1}{4}g$$

(c) (i) For $4 \leq t \leq 8$, solve

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = -\frac{1}{4}g$$

The integrating factor is
$$e^{\int 0.25\,\mathrm{d}t} = e^{t/4}$$

Multiplying the differential equation by $e^{t/4}$,
$$e^{t/4}\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25e^{t/4}v = -\frac{1}{4}ge^{t/4}$$

The left-hand side is
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/4}\right)$$

so
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/4}\right) = -\frac{1}{4}ge^{t/4}$$

Integrating,
$$\begin{aligned} ve^{t/4} &= \int -\frac{1}{4}ge^{t/4}\,\mathrm{d}t \\ &= -ge^{t/4} + C \end{aligned}$$

Hence
$$v = Ce^{-t/4} - g$$

The cage comes to rest at $t = 8$, so $v(8) = 0$. Therefore
$$\begin{aligned} 0 &= Ce^{-8/4} - g \\ 0 &= Ce^{-2} - g \\ C &= ge^2 \end{aligned}$$

Substituting for C,
$$\begin{aligned} v &= ge^2e^{-t/4} - g \\ &= g\left(e^{2-t/4} - 1\right) \end{aligned}$$

So, for $4 \le t \le 8$,
$$v = g\left(e^{2-t/4} - 1\right)$$

(ii) Using the result from part (i),
$$\begin{aligned} v(4) &= g\left(e^{2-4/4} - 1\right) \\ &= g\left(e^1 - 1\right) \\ &= g(e-1) \end{aligned}$$

With $g = 9.8$,
$$v(4) \approx 9.8(e-1) \approx 16.8\,\mathrm{m\,s^{-1}}$$

So the velocity when $t = 4$ is $g(e-1)\,\mathrm{m\,s^{-1}} \approx 16.8\,\mathrm{m\,s^{-1}}$.

(d) For $0 \le t \le 4$, the upward tension is $m(g + \lambda t)$.

Using Newton's second law with upwards positive,
$$\begin{aligned} m\frac{\mathrm{d}v}{\mathrm{d}t} &= m(g+\lambda t) - mg - 0.25mv \\ &= m\lambda t - 0.25mv \end{aligned}$$

Dividing by m,
$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = \lambda t$$

(e) (i) For $0 \le t \le 4$, solve

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = \lambda t$$

The integrating factor is

$$e^{\int 0.25\,\mathrm{d}t} = e^{t/4}$$

Multiplying through by $e^{t/4}$,

$$e^{t/4}\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25e^{t/4}v = \lambda t e^{t/4}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(ve^{t/4}\right) = \lambda t e^{t/4}$$

Integrating,

$$ve^{t/4} = \lambda \int te^{t/4}\,\mathrm{d}t + C$$

Evaluate the integral by parts. Let

$$u = t, \qquad \mathrm{d}w = e^{t/4}\,\mathrm{d}t$$

so that

$$\mathrm{d}u = \mathrm{d}t, \qquad w = 4e^{t/4}$$

Then

$$\begin{aligned}\int te^{t/4}\,\mathrm{d}t &= 4te^{t/4} - \int 4e^{t/4}\,\mathrm{d}t \\ &= 4te^{t/4} - 16e^{t/4}\end{aligned}$$

Therefore

$$ve^{t/4} = \lambda\left(4te^{t/4} - 16e^{t/4}\right) + C$$

Dividing by $e^{t/4}$,

$$v = 4\lambda(t-4) + Ce^{-t/4}$$

The cage starts from rest, so $v(0) = 0$. Hence

$$\begin{aligned}0 &= 4\lambda(0-4) + C \\ 0 &= -16\lambda + C \\ C &= 16\lambda\end{aligned}$$

Substituting back,

$$\begin{aligned}v &= 4\lambda(t-4) + 16\lambda e^{-t/4} \\ &= 4\lambda\left(t - 4 + 4e^{-t/4}\right)\end{aligned}$$

So, for $0 \le t \le 4$,

$$v = 4\lambda\left(t - 4 + 4e^{-t/4}\right)$$

(ii) The velocity is continuous at $t = 4$, so the expressions from parts (c)(ii) and (e)(i) must be equal. From part (e)(i),

$$\begin{aligned}v(4) &= 4\lambda\left(4 - 4 + 4e^{-1}\right) \\ &= \frac{16\lambda}{e}\end{aligned}$$

From part (c)(ii),

$$v(4) = g(e-1)$$

Therefore

$$\begin{aligned}\frac{16\lambda}{e} &= g(e-1)\\ \lambda &= \frac{ge(e-1)}{16}\\ &= \frac{g(e^2-e)}{16}\end{aligned}$$

With $g = 9.8$,

$$\lambda \approx \frac{9.8 \times e(e-1)}{16} \approx 2.86$$

So

$$\lambda = \frac{ge(e-1)}{16} \approx 2.86$$

(f) The total distance risen is

$$s = \int_0^4 v \,\mathrm{d}t + \int_4^8 v \,\mathrm{d}t$$

Using the expressions found for v,

$$\begin{aligned}s_1 &= \int_0^4 4\lambda \left(t - 4 + 4e^{-t/4}\right) \,\mathrm{d}t\\ &= 4\lambda \left(\int_0^4 t \,\mathrm{d}t - \int_0^4 4 \,\mathrm{d}t + \int_0^4 4e^{-t/4} \,\mathrm{d}t\right)\\ &= 4\lambda \left(\left[\frac{t^2}{2}\right]_0^4 - [4t]_0^4 + \left[-16e^{-t/4}\right]_0^4\right)\\ &= 4\lambda \left(8 - 16 + \left(-\frac{16}{e} + 16\right)\right)\\ &= 4\lambda \left(8 - \frac{16}{e}\right)\\ &= \left(32 - \frac{64}{e}\right)\lambda\end{aligned}$$

For the second stage,

$$\begin{aligned}s_2 &= \int_4^8 g\left(e^{2-t/4} - 1\right) \,\mathrm{d}t\\ &= g\left(\int_4^8 e^{2-t/4} \,\mathrm{d}t - \int_4^8 1 \,\mathrm{d}t\right)\\ &= g\left(\left[-4e^{2-t/4}\right]_4^8 - [t]_4^8\right)\\ &= g\left((-4+4e) - 4\right)\\ &= 4g(e-2)\end{aligned}$$

So

$$\begin{aligned}s &= s_1 + s_2\\ &= \left(32 - \frac{64}{e}\right)\lambda + 4g(e-2)\end{aligned}$$

Substitute $\lambda = \dfrac{ge(e-1)}{16}$:

$$\begin{aligned}
s &= \left(32 - \frac{64}{e}\right)\frac{ge(e-1)}{16} + 4g(e-2) \\
&= \left(2 - \frac{4}{e}\right)ge(e-1) + 4g(e-2) \\
&= (2e-4)\,g(e-1) + 4g(e-2) \\
&= 2g(e-2)(e-1) + 4g(e-2) \\
&= 2g(e-2)(e+1) \\
&= 2g(e^2 - e - 2)
\end{aligned}$$

Now use $g = 9.8$:

$$\begin{aligned}
s &\approx 2(9.8)(e^2 - e - 2) \\
&\approx 52.3\,\mathrm{m}
\end{aligned}$$

Therefore the total distance risen, to the nearest metre, is $52\,\mathrm{m}$.

10. A short filling pulse in a buffer tank is to be modelled. The volume of liquid in the tank at time t seconds is V litres. The pulse starts when $t = 0$, and initially the tank is empty. After 1 second the volume is measured to be 4 litres. While the pulse is occurring, V is modelled by the differential equation

$$(3t - t^2)\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{(3t - t^2)^2}{1 + (t-1)^2} + (3 - 2t)V$$

(a) By using an integrating factor show that, according to the model, while the pulse is occurring,

$$V = (3t - t^2)\left(\tan^{-1}(t-1) + 2\right)$$ **[6]**

The pulse ends when the tank is again empty.

(b) Determine the length of time that the pulse lasts according to the model **[2]**

(c) Determine, according to the model, the rate of increase of the volume at the start of the pulse. Give your answer in an exact form **[3]**

Solution

(a) First rearrange the differential equation into linear form:

$$\frac{\mathrm{d}V}{\mathrm{d}t} - \frac{3 - 2t}{3t - t^2}V = \frac{3t - t^2}{1 + (t-1)^2}$$

The integrating factor is

$$\mu(t) = e^{\int -\frac{3-2t}{3t-t^2}\,\mathrm{d}t}$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}t}(3t - t^2) = 3 - 2t$$

we have

$$\int -\frac{3 - 2t}{3t - t^2}\,\mathrm{d}t = -\ln(3t - t^2)$$

Hence

$$\mu(t) = e^{-\ln(3t - t^2)} = \frac{1}{3t - t^2}$$

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{3t - t^2}\frac{\mathrm{d}V}{\mathrm{d}t} - \frac{3 - 2t}{(3t - t^2)^2}V = \frac{1}{1 + (t-1)^2}$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{V}{3t - t^2}\right)$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{V}{3t - t^2}\right) = \frac{1}{1 + (t-1)^2}$$

Integrating with respect to t,

$$\begin{aligned}\frac{V}{3t - t^2} &= \int \frac{1}{1 + (t-1)^2}\,\mathrm{d}t \\ &= \tan^{-1}(t-1) + C\end{aligned}$$

Use the given condition $V = 4$ when $t = 1$:

$$\begin{aligned}\frac{4}{3(1) - 1^2} &= \tan^{-1}(1 - 1) + C \\ \frac{4}{2} &= 0 + C \\ C &= 2\end{aligned}$$

Therefore

$$\frac{V}{3t - t^2} = \tan^{-1}(t - 1) + 2$$

and so

$$V = (3t - t^2)\left(\tan^{-1}(t - 1) + 2\right)$$

This is the required result.

(b) The pulse ends when the tank is empty again, so set $V = 0$:

$$(3t - t^2)\left(\tan^{-1}(t - 1) + 2\right) = 0$$

Now

$$-\frac{\pi}{2} < \tan^{-1}(t - 1) < \frac{\pi}{2}$$

so

$$2 - \frac{\pi}{2} < \tan^{-1}(t - 1) + 2 < 2 + \frac{\pi}{2}$$

Since $2 - \frac{\pi}{2} > 0$, the factor $\tan^{-1}(t - 1) + 2$ is never zero. Hence

$$3t - t^2 = 0$$

so

$$\begin{aligned}t(3 - t) &= 0 \\ t &= 0 \text{ or } t = 3\end{aligned}$$

The pulse starts at $t = 0$, so it ends when $t = 3$. Therefore the pulse lasts 3 seconds.

(c) Differentiate the expression for V:

$$\begin{aligned}\frac{\mathrm{d}V}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[(3t - t^2)\left(\tan^{-1}(t - 1) + 2\right)\right] \\ &= (3 - 2t)\left(\tan^{-1}(t - 1) + 2\right) + (3t - t^2)\left(\frac{1}{1 + (t - 1)^2}\right)\end{aligned}$$

At the start of the pulse, $t = 0$:

$$\begin{aligned}\left.\frac{\mathrm{d}V}{\mathrm{d}t}\right|_{t=0} &= 3\left(\tan^{-1}(-1) + 2\right) + 0 \\ &= 3\left(-\frac{\pi}{4} + 2\right) \\ &= 6 - \frac{3\pi}{4}\end{aligned}$$

Therefore the rate of increase of volume at the start is $6 - \frac{3\pi}{4}$ L s^{-1}.

11. A horticultural mixing tank initially contains 600 litres of pure water.

A fertiliser solution flows into the tank at a constant rate of 6 litres per minute. The incoming solution contains 2 grams of fertiliser in every litre of solution.

At the same time, liquid is pumped out of the well-stirred tank at a constant rate of 9 litres per minute.

It is assumed that the fertiliser instantly dissolves uniformly throughout the tank.

Given that there are x grams of fertiliser in the tank after t minutes,

(a) Show that, while the tank still contains liquid, the situation can be modelled by the differential equation

$$\frac{dx}{dt} = 12 - \frac{3x}{200 - t}$$

[4]

(b) Hence determine after how many minutes the concentration of fertiliser in the tank first reaches 1.5 grams per litre. **[5]**

(c) Explain briefly one way in which the model could be refined. **[1]**

Solution

(a) Let V be the volume of liquid in the tank after t minutes.

Since liquid enters at 6 L/min and leaves at 9 L/min, the net change in volume is

$$\begin{aligned} V(t) &= 600 + (6 - 9)t \\ &= 600 - 3t \\ &= 3(200 - t) \end{aligned}$$

This is valid while the tank still contains liquid, so

$$600 - 3t > 0 \quad \Rightarrow \quad t < 200$$

The fertiliser enters at

$$6 \times 2 = 12 \text{ g/min}$$

The concentration in the tank at time t is

$$\frac{x}{V} = \frac{x}{600 - 3t} \text{ g/L}$$

So the fertiliser leaves at the rate

$$\begin{aligned} 9 \times \frac{x}{600 - 3t} &= \frac{9x}{600 - 3t} \\ &= \frac{9x}{3(200 - t)} \\ &= \frac{3x}{200 - t} \end{aligned}$$

Hence,

$$\begin{aligned} \frac{dx}{dt} &= \text{rate in} - \text{rate out} \\ &= 12 - \frac{3x}{200 - t} \end{aligned}$$

So the model is

$$\frac{dx}{dt} = 12 - \frac{3x}{200 - t}, \qquad t < 200$$

(b) From part (a),

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3}{200-t}x = 12$$

This is a linear differential equation. Its integrating factor is

$$\mu(t) = \exp\left(\int \frac{3}{200-t}\,\mathrm{d}t\right) = \exp\left(-3\ln(200-t)\right) = (200-t)^{-3}$$

Multiplying the differential equation by $(200-t)^{-3}$ gives

$$\begin{aligned}(200-t)^{-3}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{3x}{200-t}(200-t)^{-3} &= 12(200-t)^{-3}\\ \frac{\mathrm{d}}{\mathrm{d}t}\left(x(200-t)^{-3}\right) &= 12(200-t)^{-3}\end{aligned}$$

Integrating,

$$\begin{aligned}x(200-t)^{-3} &= \int 12(200-t)^{-3}\,\mathrm{d}t\\ &= 6(200-t)^{-2} + C\end{aligned}$$

Therefore,

$$\begin{aligned}x &= \left(6(200-t)^{-2} + C\right)(200-t)^3\\ &= 6(200-t) + C(200-t)^3\end{aligned}$$

Initially the tank contains pure water, so $x = 0$ when $t = 0$. Substituting $t = 0$ gives

$$\begin{aligned}0 &= 6(200) + C(200)^3\\ 0 &= 1200 + 8{,}000{,}000C\\ C &= -\frac{1200}{8{,}000{,}000}\\ &= -\frac{3}{20000}\end{aligned}$$

Hence

$$x = 6(200-t) - \frac{3}{20000}(200-t)^3$$

The volume at time t is

$$V = 3(200-t)$$

So the concentration is

$$\begin{aligned}\frac{x}{V} &= \frac{6(200-t) - \frac{3}{20000}(200-t)^3}{3(200-t)}\\ &= 2 - \frac{(200-t)^2}{20000}\end{aligned}$$

We want this concentration to be 1.5 g/L, so

$$\begin{aligned}2 - \frac{(200-t)^2}{20000} &= 1.5\\ \frac{(200-t)^2}{20000} &= 0.5\\ (200-t)^2 &= 10000\end{aligned}$$

Since $t < 200$, we must have $200 - t > 0$. Therefore

$$200 - t = 100$$

So

$$t = 100$$

Therefore the concentration first reaches 1.5 g/L after 100 minutes.

(c) One refinement would be to remove the assumption of perfect instantaneous mixing, and allow the fertiliser concentration to vary in different parts of the tank.

So a more realistic model could account for imperfect mixing.

12. A mixing tank initially contains 2 litres of solution in which 4 grams of dye are dissolved.
A dye solution containing 10 grams of dye per litre is pumped into the tank at a constant rate of r litres per minute and mixes instantly and uniformly with the liquid already in the tank.
At the same time, liquid is removed from the tank at a constant rate of s litres per minute.
After t minutes, the tank contains m grams of dye.

(a) (i) Show that the concentration of dye in the tank after t minutes is

$$\frac{m}{2+(r-s)t}$$

grams per litre. **[2]**

(ii) Hence show that

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{sm}{2+(r-s)t} = 10r$$ **[2]**

(b) First, consider the case where $s = r$.

(i) Solve the differential equation to find m in terms of r and t. **[4]**

(ii) Given that after 2 minutes the concentration of dye in the tank is 8 grams per litre, find r. **[2]**

(c) Now consider the case where $s = 2r$.

(i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{2}{r}$. **[1]**

(ii) Find the maximum mass of dye in the tank. **[9]**

Solution

(a) (i) Initially the tank contains 2 litres.
In t minutes, liquid enters at r litres per minute, so volume added is rt litres.
In the same time, liquid leaves at s litres per minute, so volume removed is st litres.
Hence the volume in the tank after t minutes is

$$2 + rt - st = 2 + (r-s)t$$

Since the mass of dye is m grams, the concentration is

$$\frac{m}{2+(r-s)t}$$

So the concentration of dye after t minutes is $\dfrac{m}{2+(r-s)t}$ grams per litre.

(ii) The rate at which dye enters the tank is

$$10 \times r = 10r$$

grams per minute.
From part (i), the concentration in the tank is $\dfrac{m}{2+(r-s)t}$ grams per litre, so the rate at which dye leaves is

$$s \cdot \frac{m}{2+(r-s)t}$$

grams per minute.
Therefore

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \text{rate in} - \text{rate out}$$

so

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 10r - \frac{sm}{2+(r-s)t}$$

Rearranging gives

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{sm}{2+(r-s)t} = 10r$$

Hence,

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{sm}{2+(r-s)t} = 10r$$

(b) (i) If $s = r$, then part (a)(ii) becomes

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{rm}{2+(r-r)t} = 10r$$

so

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{r}{2}m = 10r$$

Also, initially there are 4 grams of dye, so

$$m(0) = 4$$

This is a linear differential equation. Its integrating factor is

$$\exp\left(\int \frac{r}{2}\ \mathrm{d}t\right) = e^{rt/2}$$

Multiplying the differential equation by $e^{rt/2}$ gives

$$e^{rt/2}\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{r}{2}e^{rt/2}m = 10re^{rt/2}$$

The left-hand side is the derivative of $me^{rt/2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(me^{rt/2}\right) = 10re^{rt/2}$$

Integrating,

$$\begin{aligned} me^{rt/2} &= \int 10re^{rt/2}\ \mathrm{d}t \\ &= 10r \cdot \frac{2}{r}e^{rt/2} + C \\ &= 20e^{rt/2} + C \end{aligned}$$

Therefore

$$m = 20 + Ce^{-rt/2}$$

Using $m(0) = 4$,

$$\begin{aligned} 4 &= 20 + Ce^{0} \\ 4 &= 20 + C \\ C &= -16 \end{aligned}$$

Hence

$$m = 20 - 16e^{-rt/2}$$

So,

$$m = 20 - 16e^{-rt/2}$$

(ii) When $s = r$, the volume is constant and equal to

$$2 + (r - r)t = 2$$

So the concentration is

$$\frac{m}{2} = \frac{20 - 16e^{-rt/2}}{2} = 10 - 8e^{-rt/2}$$

After 2 minutes, the concentration is 8 grams per litre, so

$$8 = 10 - 8e^{-r}$$

Solving,

$$\begin{aligned} 8e^{-r} &= 2 \\ e^{-r} &= \frac{1}{4} \\ -r &= \ln\left(\frac{1}{4}\right) \\ r &= \ln 4 \end{aligned}$$

Therefore

$$r = \ln 4 \approx 1.39$$

So the required inflow rate is $r = \ln 4$ litres per minute.

(c) (i) If $s = 2r$, then the volume after t minutes is

$$2 + (r - 2r)t = 2 - rt$$

At

$$t = \frac{2}{r}$$

this volume is 0, so the tank is empty.
For $t \geq \frac{2}{r}$, the denominator in the differential equation is 0 or negative, so the model is no longer physically valid.
Hence the differential equation is not valid for $t \geq \frac{2}{r}$ because the tank is empty at $t = \frac{2}{r}$.

(ii) With $s = 2r$, the differential equation from part (a)(ii) is

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{2rm}{2 + (r - 2r)t} = 10r$$

so

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{2r}{2 - rt}m = 10r$$

with

$$m(0) = 4$$

We solve this for $0 \leq t < \frac{2}{r}$.
The integrating factor is

$$\begin{aligned} \exp\left(\int \frac{2r}{2 - rt}\,\mathrm{d}t\right) &= \exp\left(-2\ln(2 - rt)\right) \\ &= (2 - rt)^{-2} \end{aligned}$$

Multiplying through by $(2 - rt)^{-2}$ gives

$$(2 - rt)^{-2}\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{2r}{2 - rt}(2 - rt)^{-2}m = 10r(2 - rt)^{-2}$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(m(2 - rt)^{-2}\right)$$

So
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(m(2-rt)^{-2}\right) = 10r(2-rt)^{-2}$$
Integrating,
$$\begin{aligned} m(2-rt)^{-2} &= \int 10r(2-rt)^{-2}\ \mathrm{d}t \\ &= \frac{10}{2-rt} + C \end{aligned}$$
Therefore
$$m = 10(2-rt) + C(2-rt)^2$$
Using $m(0) = 4$,
$$\begin{aligned} 4 &= 10(2) + C(2)^2 \\ 4 &= 20 + 4C \\ 4C &= -16 \\ C &= -4 \end{aligned}$$
Hence
$$m = 10(2-rt) - 4(2-rt)^2$$
Expanding,
$$\begin{aligned} m &= 20 - 10rt - 4(4 - 4rt + r^2t^2) \\ &= 20 - 10rt - 16 + 16rt - 4r^2t^2 \\ &= 4 + 6rt - 4r^2t^2 \end{aligned}$$
This is a quadratic in t with negative coefficient of t^2, so it has a maximum at its stationary point. Differentiate:
$$\frac{\mathrm{d}m}{\mathrm{d}t} = 6r - 8r^2t$$
Set this equal to 0:
$$\begin{aligned} 6r - 8r^2t &= 0 \\ 6r &= 8r^2t \\ 6 &= 8rt \\ t &= \frac{3}{4r} \end{aligned}$$
Since $\frac{3}{4r} < \frac{2}{r}$, this occurs before the tank becomes empty, so it is valid.
Now substitute $t = \frac{3}{4r}$ into $m = 4 + 6rt - 4r^2t^2$:
$$\begin{aligned} m_{\max} &= 4 + 6r\left(\frac{3}{4r}\right) - 4r^2\left(\frac{3}{4r}\right)^2 \\ &= 4 + \frac{18}{4} - 4r^2 \cdot \frac{9}{16r^2} \\ &= 4 + \frac{18}{4} - \frac{36}{16} \\ &= 4 + \frac{9}{2} - \frac{9}{4} \\ &= \frac{16}{4} + \frac{18}{4} - \frac{9}{4} \\ &= \frac{25}{4} \end{aligned}$$
Therefore the maximum mass of dye in the tank is
$$\frac{25}{4}\ \mathrm{g}$$

13. The concentration C, in suitable units, of a drug in a patient's bloodstream at time t hours after an intravenous infusion begins is to be modelled. Initially, the concentration is zero, and its long-term value is M.

(a) One simple model assumes that the rate of change of concentration is directly proportional to $M - C$.

(i) Formulate a differential equation for this model. **[1]**

(ii) Verify that $C = M(1 - e^{-kt})$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition. **[3]**

An alternative model uses, for $t > 0$, the differential equation

$$\frac{\mathrm{d}C}{\mathrm{d}t} - \frac{t+2}{t(1+t+t^2)}C = S(t),$$

where $S(t)$ is a function of t.

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$\frac{1+t+t^2}{t^2}$$

[8]

(c) Suppose that $S(t) = 0$.

(i) Show that

$$C = \frac{Mt^2}{1+t+t^2}$$

[4]

(ii) Find the time predicted by this model for the concentration to reach half its long-term value. Give your answer correct to the nearest minute. **[2]**

(d) Now suppose that

$$S(t) = \frac{3t^2e^{-t}}{1+t+t^2}$$

Show that

$$C = \frac{Mt^2 - 3t^2e^{-t}}{1+t+t^2}$$

[5]

(e) It is found that the long-term concentration is 10, and C reaches half this value after 105 minutes. Determine which of the models proposed in parts (c) and (d) is more consistent with this data. **[2]**

Solution

(a) (i) Since the rate of change is directly proportional to $M - C$,

$$\frac{\mathrm{d}C}{\mathrm{d}t} = k(M - C), \qquad k > 0$$

So the differential equation is $\dfrac{\mathrm{d}C}{\mathrm{d}t} = k(M - C)$.

(ii) Take

$$C = M(1 - e^{-kt})$$

Differentiate:

$$\begin{aligned}\frac{\mathrm{d}C}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t}\left[M(1-e^{-kt})\right] \\ &= M\left(0-(-k)e^{-kt}\right) \\ &= Mke^{-kt}\end{aligned}$$

Also,

$$\begin{aligned}M - C &= M - M(1-e^{-kt}) \\ &= M - Me^{-0} + Me^{-kt}?\end{aligned}$$

The previous line should simplify directly as

$$\begin{aligned}M - C &= M - M(1-e^{-kt}) \\ &= M - M + Me^{-kt} \\ &= Me^{-kt}\end{aligned}$$

Hence,

$$\frac{\mathrm{d}C}{\mathrm{d}t} = Mke^{-kt} = k(Me^{-kt}) = k(M-C)$$

so the differential equation is satisfied.
For the initial condition,

$$C(0) = M(1-e^{0}) = M(1-1) = 0$$

For the long-term condition, since $k > 0$, $e^{-kt} \to 0$ as $t \to \infty$. Therefore

$$\lim_{t\to\infty} C = \lim_{t\to\infty} M(1-e^{-kt}) = M(1-0) = M$$

So $C = M(1-e^{-kt})$ satisfies the differential equation, the initial condition, and the long-term condition.

(b) Write the equation in the linear form

$$\frac{\mathrm{d}C}{\mathrm{d}t} + P(t)C = S(t)$$

where

$$P(t) = -\frac{t+2}{t(1+t+t^2)}$$

To find the integrating factor, first decompose $P(t)$ into partial fractions:

$$-\frac{t+2}{t(1+t+t^2)} = \frac{A}{t} + \frac{Bt+C}{1+t+t^2}$$

Multiplying through by $t(1+t+t^2)$ gives

$$\begin{aligned}-(t+2) &= A(1+t+t^2) + t(Bt+C) \\ &= A + At + At^2 + Bt^2 + Ct \\ &= A + (A+C)t + (A+B)t^2\end{aligned}$$

Comparing coefficients with $-2 - t + 0t^2$:

$$\begin{aligned}A &= -2 \\ A + C &= -1 \\ A + B &= 0\end{aligned}$$

So

$$A = -2, \qquad C = 1, \qquad B = 2$$

Therefore

$$P(t) = -\frac{2}{t} + \frac{2t+1}{1+t+t^2}$$

The integrating factor is

$$\mu(t) = e^{\int P(t)\ \mathrm{d}t}$$

Since $t > 0$,

$$\begin{aligned}\int P(t)\ \mathrm{d}t &= \int \left(-\frac{2}{t} + \frac{2t+1}{1+t+t^2}\right)\ \mathrm{d}t\\ &= -2\ln t + \ln(1+t+t^2)\end{aligned}$$

Hence

$$\begin{aligned}\mu(t) &= e^{-2\ln t + \ln(1+t+t^2)}\\ &= e^{\ln(1+t+t^2)}e^{-2\ln t}\\ &= (1+t+t^2)t^{-2}\\ &= \frac{1+t+t^2}{t^2}\end{aligned}$$

So the integrating factor is $\dfrac{1+t+t^2}{t^2}$.

(c) (i) When $S(t) = 0$, the differential equation becomes

$$\frac{\mathrm{d}C}{\mathrm{d}t} - \frac{t+2}{t(1+t+t^2)}C = 0$$

Using the integrating factor

$$\mu(t) = \frac{1+t+t^2}{t^2}$$

gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mu C\right) = 0$$

So

$$\mu C = A$$

where A is a constant. Thus

$$\begin{aligned}C &= \frac{A}{\mu}\\ &= \frac{A}{(1+t+t^2)/t^2}\\ &= \frac{At^2}{1+t+t^2}\end{aligned}$$

Now use the long-term value:

$$\begin{aligned}\lim_{t\to\infty} C &= \lim_{t\to\infty}\frac{At^2}{1+t+t^2}\\ &= \lim_{t\to\infty}\frac{A}{t^{-2}+t^{-1}+1}\\ &= A\end{aligned}$$

Since the long-term concentration is M, we must have $A = M$. Therefore

$$C = \frac{Mt^2}{1+t+t^2}$$

(ii) Half the long-term value is $\frac{M}{2}$. Set

$$\frac{Mt^2}{1+t+t^2} = \frac{M}{2}$$

Cancelling M and solving:

$$\begin{aligned} \frac{t^2}{1+t+t^2} &= \frac{1}{2} \\ 2t^2 &= 1+t+t^2 \\ t^2 - t - 1 &= 0 \end{aligned}$$

Hence

$$t = \frac{1 \pm \sqrt{5}}{2}$$

Since $t > 0$,

$$t = \frac{1+\sqrt{5}}{2} \text{ hours} \approx 1.618 \text{ hours}$$

Converting to minutes:

$$1.618 \times 60 \approx 97.1$$

So the concentration reaches half its long-term value after 97 minutes, to the nearest minute.

(d) Using the integrating factor from part (b),

$$\mu(t) = \frac{1+t+t^2}{t^2}$$

and

$$S(t) = \frac{3t^2e^{-t}}{1+t+t^2}$$

Then

$$\begin{aligned} \mu(t)S(t) &= \frac{1+t+t^2}{t^2} \cdot \frac{3t^2e^{-t}}{1+t+t^2} \\ &= 3e^{-t} \end{aligned}$$

So the differential equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mu C) = 3e^{-t}$$

Integrating:

$$\begin{aligned} \mu C &= \int 3e^{-t}\, \mathrm{d}t \\ &= -3e^{-t} + A \end{aligned}$$

Hence

$$\begin{aligned} C &= \frac{-3e^{-t} + A}{\mu} \\ &= \frac{t^2(-3e^{-t} + A)}{1+t+t^2} \\ &= \frac{At^2 - 3t^2e^{-t}}{1+t+t^2} \end{aligned}$$

Now use the long-term value. As $t \to \infty$,

$$\frac{At^2}{1+t+t^2} \to A$$

and, using the given fact $\lim_{t\to\infty} t^2e^{-t} = 0$,

$$\frac{3t^2e^{-t}}{1+t+t^2} \to 0$$

Therefore

$$\lim_{t\to\infty} C = A$$

Since the long-term concentration is M, $A = M$.

So

$$C = \frac{Mt^2 - 3t^2e^{-t}}{1+t+t^2}$$

as required.

(e) Here $M = 10$, so half the long-term concentration is 5, reached after 105 minutes = 1.75 hours.

From part (c), model (c) predicts the half-value time to be 97 minutes.

For model (d), at $t = 1.75$,

$$\begin{aligned} C &= \frac{10t^2 - 3t^2e^{-t}}{1+t+t^2} \\ &= \frac{10(1.75)^2 - 3(1.75)^2e^{-1.75}}{1+1.75+(1.75)^2} \\ &\approx \frac{30.625 - 1.598}{5.8125} \\ &\approx 4.99 \end{aligned}$$

This is essentially 5.

Therefore model (d) is more consistent with the data.

14. A drinks manufacturer uses a mixing tank that initially contains 4 litres of water. A flavouring liquid, which is 60% fruit concentrate by volume, is pumped into the tank at a constant rate of p litres per minute. At the same time, the contents of the tank are thoroughly mixed and drawn off at a constant rate of q litres per minute.

After t minutes, let x litres be the amount of fruit concentrate in the tank. You may assume that mixing is instantaneous and uniform.

(a) (i) Show that the proportion of fruit concentrate in the tank after t minutes is

$$\frac{x}{4+(p-q)t}$$ **[2]**

(ii) Hence show that

$$\frac{\mathrm{d}x}{\mathrm{d}t}+\frac{qx}{4+(p-q)t}=\frac{3p}{5}$$ **[2]**

(b) Now consider the case where $q = p$.

(i) Solve the differential equation to find x in terms of p and t. **[4]**

(ii) Given that after 4 minutes the liquid in the tank is 30% fruit concentrate, find the value of p. **[2]**

(c) Now consider the case where $q = 2p$.

(i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{4}{p}$. **[1]**

(ii) Find the maximum amount of fruit concentrate in the tank before the tank empties. **[9]**

Solution

(a) (i) After t minutes, the volume in the tank is

$$4+pt-qt=4+(p-q)t$$

Since there are x litres of fruit concentrate in the tank, the proportion of fruit concentrate is

$$\frac{x}{4+(p-q)t}$$

Hence the required proportion is $\dfrac{x}{4+(p-q)t}$.

(ii) The flavouring liquid entering the tank is 60% concentrate, so the rate at which concentrate enters is

$$0.6p=\frac{3p}{5}$$

From part (i), the proportion of concentrate in the tank is $\dfrac{x}{4+(p-q)t}$, so the rate at which concentrate leaves is

$$q\left(\frac{x}{4+(p-q)t}\right)=\frac{qx}{4+(p-q)t}$$

Therefore

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t}&=\text{rate in}-\text{rate out}\\&=\frac{3p}{5}-\frac{qx}{4+(p-q)t}\end{aligned}$$

Rearranging,

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{qx}{4+(p-q)t} = \frac{3p}{5}$$

Hence the differential equation is shown.

(b) (i) If $q = p$, then the differential equation from part (a)(ii) becomes

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{px}{4+(p-p)t} = \frac{3p}{5}$$

so

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{p}{4}x = \frac{3p}{5}$$

Initially the tank contains only water, so

$$x(0) = 0$$

The integrating factor is

$$\mu(t) = e^{\int \frac{p}{4}\,\mathrm{d}t} = e^{pt/4}$$

Multiplying the differential equation by $e^{pt/4}$ gives

$$\begin{aligned} e^{pt/4}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{p}{4}e^{pt/4}x &= \frac{3p}{5}e^{pt/4} \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(xe^{pt/4}\right) &= \frac{3p}{5}e^{pt/4} \end{aligned}$$

Integrating,

$$\begin{aligned} xe^{pt/4} &= \int \frac{3p}{5}e^{pt/4}\,\mathrm{d}t \\ &= \frac{3p}{5}\cdot\frac{4}{p}e^{pt/4} + C \\ &= \frac{12}{5}e^{pt/4} + C \end{aligned}$$

Hence

$$x = \frac{12}{5} + Ce^{-pt/4}$$

Using $x(0) = 0$,

$$0 = \frac{12}{5} + C$$

so

$$C = -\frac{12}{5}$$

Therefore

$$x = \frac{12}{5}\left(1 - e^{-pt/4}\right)$$

So the solution is $x = \dfrac{12}{5}\left(1 - e^{-pt/4}\right)$.

(ii) When $q = p$, the total volume stays constant at 4 litres.
After 4 minutes, the liquid is 30% concentrate, so the amount of concentrate is

$$x = 0.3 \times 4 = \frac{6}{5}$$

Substitute $t = 4$ into the formula from part (i):

$$\frac{12}{5}\left(1 - e^{-p}\right) = \frac{6}{5}$$

Multiply through by 5:

$$12(1 - e^{-p}) = 6$$

Hence

$$1 - e^{-p} = \frac{1}{2}$$
$$e^{-p} = \frac{1}{2}$$

Taking natural logarithms,

$$-p = \ln\left(\frac{1}{2}\right) = -\ln 2$$

so

$$p = \ln 2$$

Therefore $p = \ln 2 \approx 0.693$ litres min^{-1}.

(c) (i) If $q = 2p$, then the volume after t minutes is

$$4 + (p - 2p)t = 4 - pt$$

At

$$t = \frac{4}{p}$$

this volume is 0, so the tank is empty. For $t \geq \frac{4}{p}$, the denominator $4 - pt$ is zero or negative, which is not physically meaningful.
Therefore the differential equation is only valid for $0 \leq t < \frac{4}{p}$.

(ii) With $q = 2p$, the differential equation becomes

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2px}{4 - pt} = \frac{3p}{5}$$

for $0 \leq t < \frac{4}{p}$, with

$$x(0) = 0$$

The integrating factor is

$$\mu(t) = e^{\int \frac{2p}{4-pt}\,\mathrm{d}t}$$

Now let $u = 4 - pt$, so $\mathrm{d}u = -p\,\mathrm{d}t$. Then

$$\begin{aligned}\int \frac{2p}{4 - pt}\,\mathrm{d}t &= -2\int \frac{1}{u}\,\mathrm{d}u \\ &= -2\ln u \\ &= -2\ln(4 - pt)\end{aligned}$$

Hence

$$\mu(t) = e^{-2\ln(4-pt)} = (4 - pt)^{-2}$$

Multiply the differential equation by $(4 - pt)^{-2}$:

$$(4 - pt)^{-2}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{2p}{4 - pt}(4 - pt)^{-2}x = \frac{3p}{5}(4 - pt)^{-2}$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(x(4 - pt)^{-2}\right)$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(x(4 - pt)^{-2}\right) = \frac{3p}{5}(4 - pt)^{-2}$$

Integrating,

$$x(4 - pt)^{-2} = \int \frac{3p}{5}(4 - pt)^{-2}\,\mathrm{d}t$$

Again let $u = 4 - pt$, so $\mathrm{d}u = -p\,\mathrm{d}t$. Then

$$\begin{aligned}\int \frac{3p}{5}(4-pt)^{-2}\,\mathrm{d}t &= -\frac{3}{5}\int u^{-2}\,\mathrm{d}u \\ &= -\frac{3}{5}\left(\frac{u^{-1}}{-1}\right) + C \\ &= \frac{3}{5}u^{-1} + C \\ &= \frac{3}{5(4-pt)} + C\end{aligned}$$

Therefore

$$x(4-pt)^{-2} = \frac{3}{5(4-pt)} + C$$

Multiply by $(4-pt)^2$:

$$x = \frac{3}{5}(4-pt) + C(4-pt)^2$$

Use $x(0) = 0$:

$$\begin{aligned}0 &= \frac{3}{5}(4) + C(4)^2 \\ 0 &= \frac{12}{5} + 16C\end{aligned}$$

so

$$C = -\frac{3}{20}$$

Hence

$$\begin{aligned}x &= \frac{3}{5}(4-pt) - \frac{3}{20}(4-pt)^2 \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{3}{20}\left(16 - 8pt + p^2t^2\right) \\ &= \frac{12}{5} - \frac{3pt}{5} - \frac{12}{5} + \frac{6pt}{5} - \frac{3p^2t^2}{20} \\ &= \frac{3pt}{5} - \frac{3p^2t^2}{20}\end{aligned}$$

So

$$x = \frac{3pt}{5} - \frac{3p^2t^2}{20} = \frac{3}{20}pt(4-pt)$$

To find the maximum value, differentiate:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3p}{5} - \frac{3p^2t}{10}$$

Set this equal to 0:

$$\begin{aligned}\frac{3p}{5} - \frac{3p^2t}{10} &= 0 \\ \frac{3p}{10}(2-pt) &= 0\end{aligned}$$

Since $p > 0$, this gives

$$2 - pt = 0$$

so

$$t = \frac{2}{p}$$

This lies in the interval $0 \le t < \frac{4}{p}$, and since the quadratic has negative coefficient of t^2, this gives the maximum.

Now substitute $t = \frac{2}{p}$:

$$\begin{aligned} x_{\max} &= \frac{3p}{5}\left(\frac{2}{p}\right) - \frac{3p^2}{20}\left(\frac{2}{p}\right)^2 \\ &= \frac{6}{5} - \frac{3}{20} \cdot 4 \\ &= \frac{6}{5} - \frac{12}{20} \\ &= \frac{6}{5} - \frac{3}{5} \\ &= \frac{3}{5} \end{aligned}$$

Therefore the maximum amount of fruit concentrate is $\dfrac{3}{5}$ litres, occurring at $t = \dfrac{2}{p}$ minutes.